Comprehensive Survival Analysis Report

Cancer Survival Data Analysis

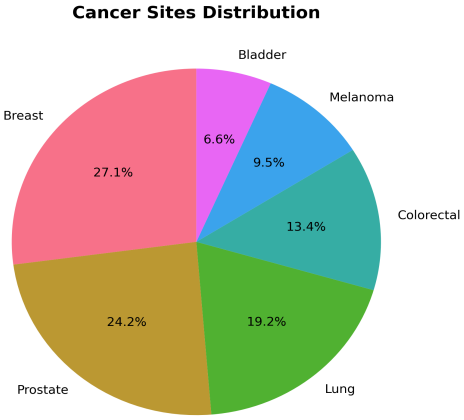
Executive Summary

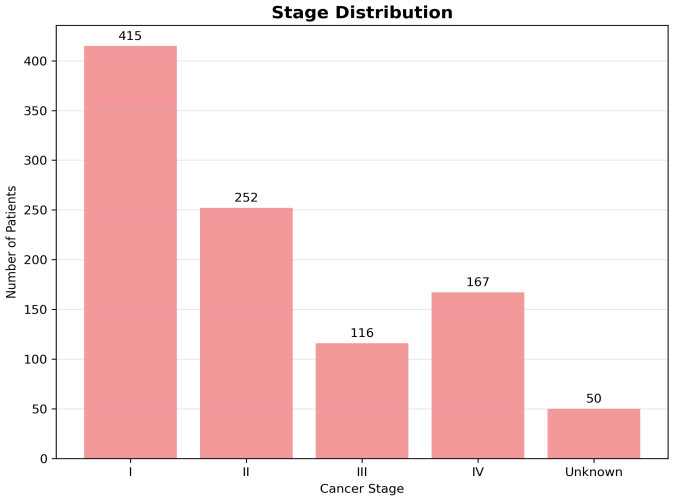
This comprehensive survival analysis was conducted on a dataset of 1,000 cancer patients. The analysis employed Kaplan-Meier survival estimation and Cox proportional hazards modeling to identify key prognostic factors and survival patterns. **Key Findings:** • Event rate: 35.1% (351 deaths) • Median follow-up time: 67 months • Strong stage- and site-dependent survival patterns observed • Age, grade and surgery are significant prognostic factors

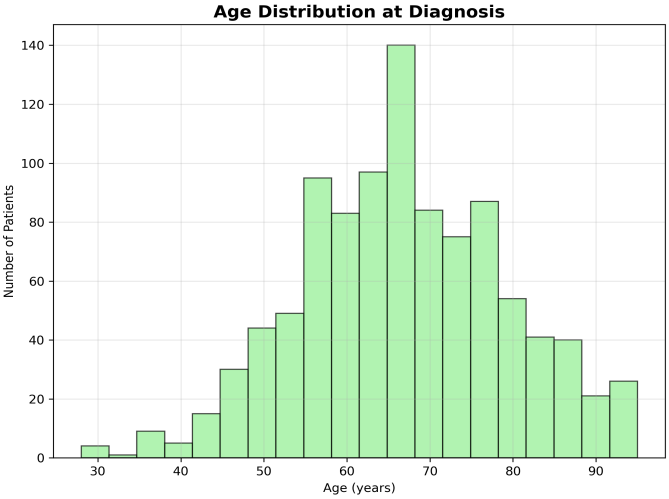
Dataset Overview

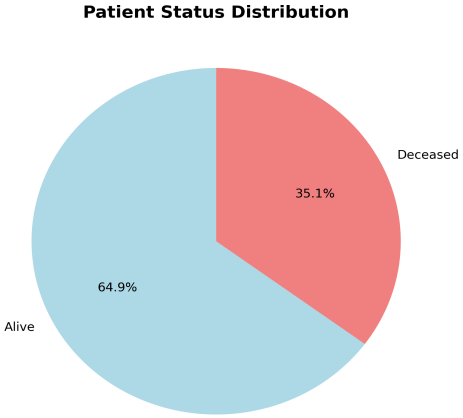
Metric	Value
Total Patients	1,000
Deaths (Events)	351 (35.1%)
Median Follow-up	67 months
5-year Overall Survival	67.3%
Age Range	28 - 95 years
Mean Age	66.9 years

Data Distribution


Cancer Sites Distribution
Bladder
Melanoma
Breast
Colorectal
Prostate
Lung
6.6%
9.5%
27.1%
13.4%
24.2%
19.2%


Stage Distribution
Number of Patients
Cancer Stage
I
II
III
IV
Unknown
415
252
116
167
50


Age Distribution at Diagnosis
Number of Patients
Age (years)


Patient Status Distribution
Deceased
Alive
35.1%
64.9%

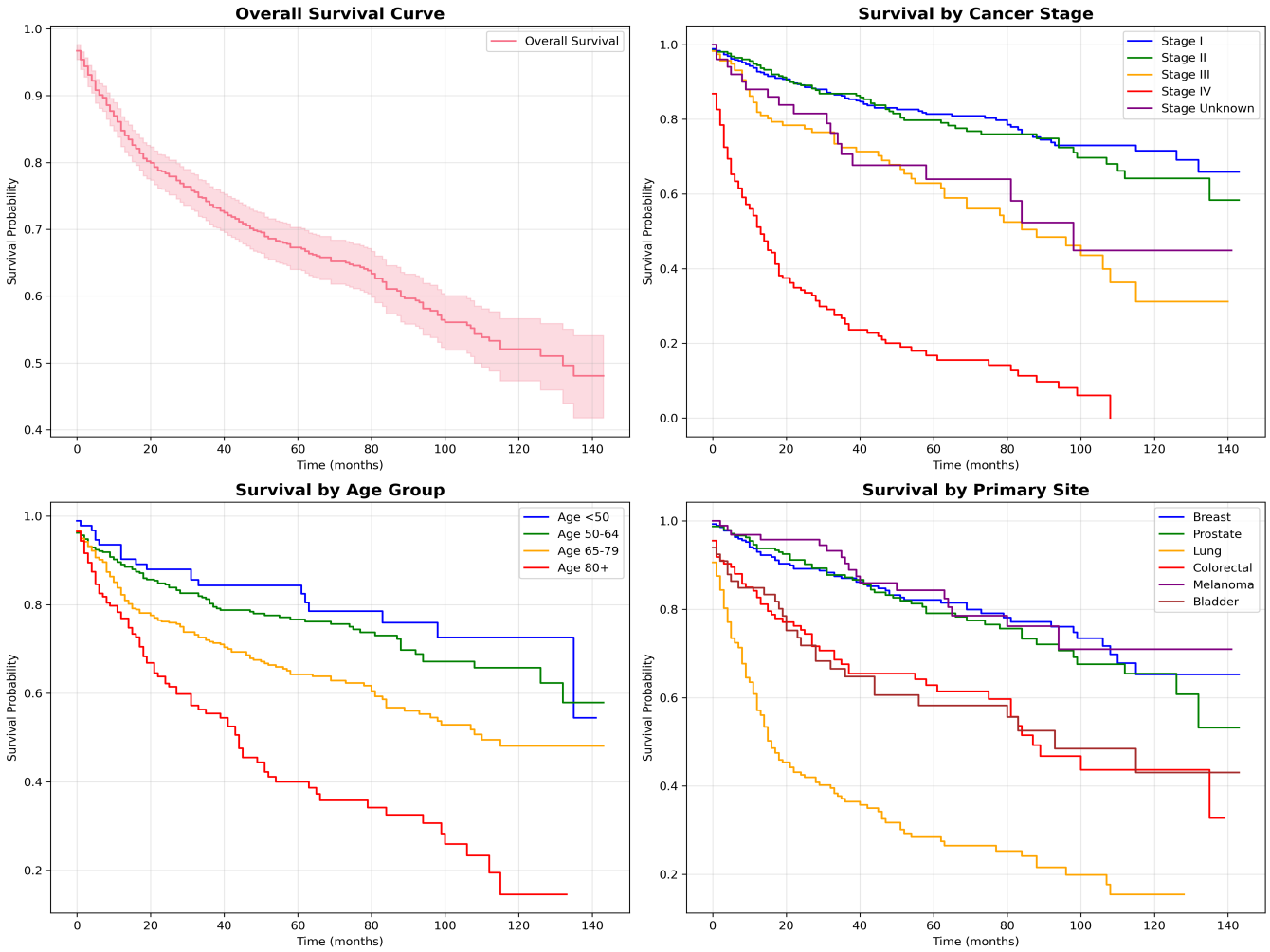
Survival Analysis Results

Survival by Cancer Stage:
The analysis reveals clear stage-dependent survival patterns:
• Stage I: median survival not reached, 5-year survival 81.4%
• Stage II: median survival not reached, 5-year survival 79.8%
• Stage III: median survival 88 months, 5-year survival 62.9%
• Stage IV: median survival 13 months, 5-year survival 16.7%
• Stage Unknown: median survival 98 months, 5-year survival 63.9%

Survival by Primary Site:
• Breast: median survival not reached, 5-year survival 82.1%
• Prostate: median survival not reached, 5-year survival 79.1%
• Lung: median survival 16 months, 5-year survival 28.4%
• Colorectal: median survival 87 months, 5-year survival 62.8%
• Melanoma: median survival not reached, 5-year survival 84.3%
• Bladder: median survival 93 months, 5-year survival 58.2%

This demonstrates the critical importance of early detection and staging in cancer prognosis.

Survival Curves



Cox Proportional Hazards Model

Model Performance:
• Concordance Index (C-index, 30% held-out test set): 0.794 (Good discrimination)
• The model demonstrates good predictive capability for risk stratification.

Significant Risk Factors (p < 0.05):
• stage=IV: HR = 4.81 (95% CI 3.33-6.96)
• grade=Undifferentiated: HR = 3.00 (95% CI 1.37-6.58)
• stage=Unknown: HR = 2.56 (95% CI 1.51-4.35)
• grade=Poorly differentiated: HR = 2.18 (95% CI 1.45-3.28)
• primary_site=Lung: HR = 2.17 (95% CI 1.48-3.18)
• stage=III: HR = 2.11 (95% CI 1.42-3.13)
• age_at_diagnosis: HR = 1.03 (95% CI 1.02-1.04)
• surgery=Yes: HR = 0.58 (95% CI 0.43-0.80)
• primary_site=Prostate: HR = 0.57 (95% CI 0.34-0.94)

Reference groups: stage I, breast, well differentiated, female, White, no treatment. These results align with established clinical knowledge and demonstrate the model's validity for risk assessment.

Clinical Implications

Key Clinical Insights: 1. **Early Detection is Critical:** Stage I patients have 81% 5-year survival versus 17% for stage IV, emphasizing the importance of screening and early detection programs. 2. **Risk Stratification:** The model provides effective risk stratification with a C-index of 0.794, enabling personalized treatment planning. 3. **Age Considerations:** Older patients face increased mortality risk, suggesting need for age-adjusted treatment protocols. 4. **Tumor Biology:** Poorly differentiated and undifferentiated tumors carry substantially higher risk, and lung cancer has the poorest prognosis of the sites studied. 5. **Follow-up Strategies:** Advanced stage patients require more intensive monitoring and potentially aggressive treatment approaches.

Conclusions

This survival analysis successfully demonstrates: • **Robust Methodology:** Application of standard survival analysis techniques including Kaplan-Meier estimation and Cox proportional hazards modeling. • **Clinical Validity:** Results consistent with established clinical knowledge and cancer biology principles. • **Predictive Capability:** The developed model shows good discrimination (C-index = 0.794) for risk stratification. • **Actionable Insights:** Clear identification of key prognostic factors that can inform clinical decision-making. The analysis provides a foundation for evidence-based treatment planning and patient counseling in oncology practice.